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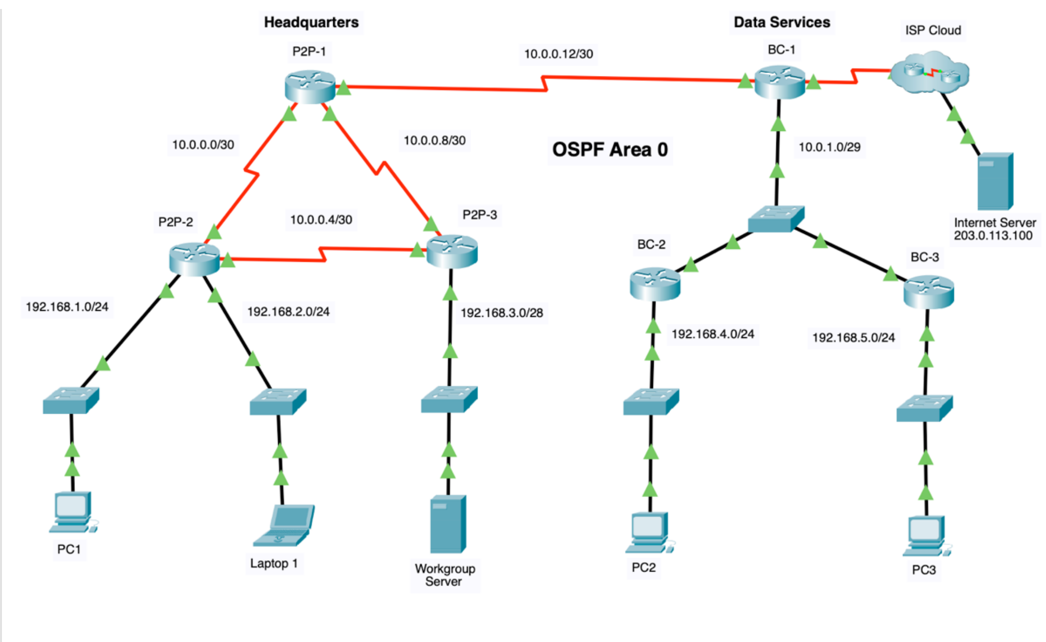
รายงานผลการปฏิบัติการ: 2.7.1 Packet Tracer – Single-Area OSPFv2 Configuration (Lab 13)

วัตถุประสงค์ของการทดลอง

- ติดตั้งและกำหนดค่า Single-Area OSPFv2 (Area 0) บนเครือข่ายแบบ Point-to-Point และ Broadcast Multiaccess
- กำหนดค่า Router ID บนเราเตอร์ในเครือข่าย Multiaccess
- เปิดใช้งาน OSPFv2 ผ่านคำสั่ง network ร่วมกับ Wildcard Mask และการเปิดใช้งานผ่าน Interface โดยตรง
- ปรับแต่ง OSPF เพื่อเพิ่มประสิทธิภาพและความปลอดภัย (Passive Interface, OSPF Priority, Cost, Reference Bandwidth และ Timers)
- สร้างและกระจายเส้นทาง Default Route ไปยังเราเตอร์ทุกตัวในเครือข่าย OSPF



ตารางแสดงการกำหนดค่า IP Address (Addressing Table)

อุปกรณ์	Interface	IP Address / Prefix	เชื่อมต่อไปยัง
P2P-1	S0/1/0	10.0.0.1/30	P2P-2 (S0/1/0)
	S0/1/1	10.0.0.9/30	P2P-3 (S0/1/1)

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	S0/2/0	10.0.0.13/30	BC-1 (S0/1/0)
P2P-2	S0/1/0	10.0.0.2/30	P2P-1 (S0/1/0)
	S0/1/1	10.0.0.5/30	P2P-3 (S0/1/0)
	G0/0/0	192.168.1.1/24	PC 1 LAN
	G0/0/1	192.168.2.1/24	Laptop 1 LAN
P2P-3	S0/1/0	10.0.0.6/30	P2P-2 (S0/1/1)
	S0/1/1	10.0.0.10/30	P2P-1 (S0/1/1)
	G0/0/0	192.168.3.1/28	Workgroup Server LAN
BC-1	S0/1/0	10.0.0.14/30	P2P-1 (S0/2/0)
	S0/1/1	64.0.100.2/30	ISP Cloud
	G0/0/0	10.0.1.1/29	Central Switch (Multiaccess)
BC-2	G0/0/0	192.168.4.1/24	PC 2 LAN
	G0/0/1	10.0.1.2/29	Central Switch (Multiaccess)
BC-3	G0/0/0	192.168.5.1/24	PC 3 LAN
	G0/0/1	10.0.1.3/29	Central Switch (Multiaccess)
Internet Server	NIC	203.0.113.100/24	ISP Cloud

สรุปการกำหนดค่าตามข้อกำหนด (Implementation Details)

1. การเปิดใช้งาน OSPF Process ID 10 และประกาศ Network (Headquarters)

- à¸□à¸±à¹ˆà¸‡à¹€à¸„à¸£à¸·à¸à¸‚à¹ˆà¸²à¸¢ HQ (P2P-1, P2P-2, P2P-3) à¹ƒà¸Šà¹‰à¸„à¸³à¸ªà¸±à¹ˆà¸‡ network area 0 à¸ à¸²à¸¢à¹ƒà¸•à¹‰à¹‚à¸«à¸¡à¸” router ospf 10

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- ตัวอย่างการแปลง Subnet Mask เป็น Inverse (Wildcard) Mask:
    - /30 (255.255.255.252) \rightarrow 0.0.0.3
    - /28 (255.255.255.240) \rightarrow 0.0.0.15
    - /24 (255.255.255.0) \rightarrow 0.0.0.255

2. การเปิดใช้งาน OSPF ระดับ Interface และ Router ID (Data Services)

- ฝั่ง Data Service (BC-1, BC-2, BC-3) ใช้คำสั่ง ip ospf 10 area 0 บน Interface ที่ต้องการโดยตรง
- กำหนดค่า Router ID เพื่อเจาะจงตัวตนบนเครือข่าย Multiaccess:
    - **BC-1:** 6.6.6.6
    - **BC-2:** 5.5.5.5
    - **BC-3:** 4.4.4.4

3. การควบคุม Traffic และความปลอดภัย (Passive Interface & Default Route)

- ตั้งค่า passive-interface บน Interface ที่เชื่อมต่อไปยัง LAN ของ Client/Server (เช่น G0/0/0, G0/0/1) และลิงก์ที่ไปยัง ISP เพื่อหยุดการส่ง OSPF Hello Packets ออกสู่อุปกรณ์ปลายทาง
- กำหนด Static Default Route บน **BC-1**: ip route 0.0.0.0 0.0.0.0 Serial0/1/1
- แจกจ่าย Default Route ไปยังเครือข่าย OSPF ทั้งหมดด้วยคำสั่ง default-information originate ภายใต้ OSPF Process บน BC-1

4. การปรับแต่ง OSPF Path Metric และการเลือก DR

- **Auto-Cost Reference-Bandwidth:** ตั้งค่า auto-cost reference-bandwidth 1000 บนเราเตอร์ทุกตัว เพื่อให้ Cost ของ Gigabit Ethernet เท่ากับ 1 (1000 / 1000) และ Fast Ethernet เท่ากับ 10 (1000 / 100)
- **Manual Cost:** ปรับแต่ง Cost บนพอร์ต Serial0/1/1 ของ **P2P-1** ให้เป็น 50 (ip ospf cost 50)
- **DR Election:** ตั้งค่า ip ospf priority 255 บนพอร์ต G0/0/0 ของ **BC-1** เพื่อการันตีให้ BC-1 ได้รับเลือกเป็น Designated Router (DR) ตลอดเวลา
- **Timers:** ปรับค่า Hello/Dead Timers เป็นสองเท่า (Hello 20 วินาที, Dead 80 วินาที) บนพอร์ตเชื่อมต่อระหว่าง **P2P-1** (S0/2/0) และ **BC-1** (S0/1/0)

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ชุดคำสั่ง Configuration ที่สำคัญของแต่ละอุปกรณ์

P2P-1

```
interface Serial0/1/1
 ip ospf cost 50
interface Serial0/2/0
 ip ospf hello-interval 20
 ip ospf dead-interval 80
router ospf 10
 auto-cost reference-bandwidth 1000
 network 10.0.0.0 0.0.0.3 area 0
 network 10.0.0.8 0.0.0.3 area 0
 network 10.0.0.12 0.0.0.3 area 0
```

P2P-2

```
router ospf 10
 auto-cost reference-bandwidth 1000
 passive-interface GigabitEthernet0/0/0
 passive-interface GigabitEthernet0/0/1
 network 10.0.0.0 0.0.0.3 area 0
 network 10.0.0.4 0.0.0.3 area 0
 network 192.168.1.0 0.0.0.255 area 0
 network 192.168.2.0 0.0.0.255 area 0
```

P2P-3

```
router ospf 10
 auto-cost reference-bandwidth 1000
 passive-interface GigabitEthernet0/0/0
 network 10.0.0.4 0.0.0.3 area 0
```

```
 network 10.0.0.8 0.0.0.3 area 0
 network 192.168.3.0 0.0.0.15 area 0
```

BC-1

```
interface GigabitEthernet0/0/0
 ip ospf priority 255
 ip ospf 10 area 0
interface Serial0/1/0
 ip ospf hello-interval 20
 ip ospf dead-interval 80
 ip ospf 10 area 0
router ospf 10
 router-id 6.6.6.6
 passive-interface Serial0/1/1
 auto-cost reference-bandwidth 1000
 default-information originate
ip route 0.0.0.0 0.0.0.0 Serial0/1/1
```

BC-2

```
interface GigabitEthernet0/0/0
 ip ospf 10 area 0
interface GigabitEthernet0/0/1
 ip ospf 10 area 0
router ospf 10
 router-id 5.5.5.5
 passive-interface GigabitEthernet0/0/0
 auto-cost reference-bandwidth 1000
```

BC-3

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```
interface GigabitEthernet0/0/0
 ip ospf 10 area 0
interface GigabitEthernet0/0/1
 ip ospf 10 area 0
router ospf 10
 router-id 4.4.4.4
 passive-interface GigabitEthernet0/0/0
 auto-cost reference-bandwidth 1000
```

ผลการตรวจสอบและการทดสอบ (Verification & Testing)

- **OSPF Adjacencies:** ตรวจสอบผ่าน show ip ospf neighbor พบว่าเราเตอร์ทุกตัวสร้าง Neighbor Adjacency แบบ FULL ได้สมบูรณ์ โดยบนฝั่ง Multiaccess Switch (G0/0/0) เราเตอร์ BC-1 ทำหน้าที่เป็น DR (Priority 255)
- **Routing Table:** ตรวจสอบผ่าน show ip route พบว่าเราเตอร์ทุกตัวเรียนรู้ Subnet ต่างๆ ภายในเครือข่าย OSPF ครบถ้วน และได้รับเส้นทาง Default Route (O*E2 0.0.0.0/0) ที่ส่งมาจาก BC-1
- **Connectivity Test:** ทดสอบ Ping จากโฮสต์ปลายทางทุกเครื่อง (PC1, Laptop1, Workgroup Server, PC2, PC3) ไปยัง Internet Server (203.0.113.100) ให้ผลลัพธ์ **Successful 100%**

การทดลองนี้แสดงให้เห็นถึงขั้นตอนการออกแบบและปรับแต่ง OSPFv2 ทั้งสองรูปแบบ ได้แก่ เครือข่าย Point-to-Point แบบ Serial ที่ไม่มีการเลือก DR/BDR และเครือข่าย Broadcast Multiaccess ผ่านสวิตช์ที่มีการควบคุมบทบาท DR ผ่านค่า Priority การใช้งานคำสั่ง passive-interface และ auto-cost reference-bandwidth ร่วมกับการปรับแต่ง Timers ทำให้ระบบเครือข่ายมีเสถียรภาพ ประมวลผลเส้นทางได้อย่างมีประสิทธิภาพ และมีความปลอดภัยสูงสุด

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Assessment Items	Status	Points	Component(s)	Feedback
Network				
BC-1				
OSPF				
Process ID 10				
Area		0	Routing	
Area 0		1	Routing	
Area Status	Correct	1	Activate OSPF p…	
Auto Cost	Correct	2	Modify OSPF O…	
Default Information	Correct	2	Propagate defau…	
Passive Interface		0	Routing	
Serial0/1/1	Correct	2	Modify OSPF O…	
Router ID	Correct	3	Configure Route…	
Ports				
GigabitEthernet0/0/0				
OSPF Area	Correct	1	Configure OSPF…	
OSPF Id	Correct	1	Configure OSPF…	
OSPF Priority	Correct	4	Modify OSPF O…	
Serial0/1/0				
OSPF Area	Correct	1	Configure OSPF…	
OSPF Dead-Interval	Correct	4	Modify OSPF O…	
OSPF Hello-Interval	Correct	4	Modify OSPF O…	
OSPF Id	Correct	1	Configure OSPF…	
Routes		0	Other	
Static Routes		0	Routing	
Route0	Correct	6	Propagate defau…	
BC-2				
OSPF				
Process ID 10				
Area		0	Routing	
Area 0		1	Routing	
Area Status	Correct	1	Activate OSPF p…	
Auto Cost	Correct	2	Modify OSPF O…	
Passive Interface		0	Routing	
GigabitEthernet0/0/0	Correct	2	Modify OSPF O…	
Router ID	Correct	3	Configure Route…	
Ports				
GigabitEthernet0/0/0				
OSPF Area	Correct	1	Configure OSPF…	
OSPF Id	Correct	1	Configure OSPF…	
GigabitEthernet0/0/1				
OSPF Area	Correct	1	Configure OSPF…	
OSPF Id	Correct	1	Configure OSPF…	
BC-3				
OSPF				
Process ID 10				
Area		0	Routing	
Area 0		1	Routing	
Area Status	Correct	1	Activate OSPF p…	
Auto Cost	Correct	2	Modify OSPF O…	
Passive Interface		0	Routing	
GigabitEthernet0/0/0	Correct	2	Modify OSPF O…	
Router ID	Correct	3	Configure Route…	
Ports				
GigabitEthernet0/0/0				
OSPF Area	Correct	1	Configure OSPF…	
OSPF Id	Correct	1	Configure OSPF…	
GigabitEthernet0/0/1				
OSPF Area	Correct	1	Configure OSPF…	
OSPF Id	Correct	1	Configure OSPF…	
P2P-1				
OSPF				
Process ID 10				
Area		0	Routing	
Area 0		1	Routing	
Area Status	Correct	1	Activate OSPF p…	
Auto Cost	Correct	2	Modify OSPF O…	
Networks				
Route0	Correct	2	Configure OSPF…	
Route1	Correct	2	Configure OSPF…	
Route2	Correct	2	Configure OSPF…	
Ports				
Serial0/1/1		0	Other	
OSPF Cost	Correct	4	Modify OSPF O…	
Serial0/2/0				
OSPF Dead-Interval	Correct	4	Modify OSPF O…	
OSPF Hello-Interval	Correct	4	Modify OSPF O…	

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OSPF Hello-Interval	Correct	4	Modify OSPF O...
P2P-2			
OSPF			
Process ID 10			
Area		0	Routing
Area 0		1	Activate OSPF p...
Area Status	Correct	1	Activate OSPF p...
Auto Cost	Correct	2	Modify OSPF O...
Networks			
Route0	Correct	2	Configure OSPF...
Route1	Correct	2	Configure OSPF...
Route2	Correct	2	Configure OSPF...
Route3	Correct	2	Configure OSPF...
Passive Interface			
GigabitEthernet0/0/0	Correct	2	Modify OSPF O...
GigabitEthernet0/0/1	Correct	2	Modify OSPF O...
P2P-3			
OSPF			
Process ID 10			
Area		0	Routing
Area 0		1	Activate OSPF p...
Area Status	Correct	1	Activate OSPF p...
Auto Cost	Correct	2	Modify OSPF O...
Networks			
Route0	Correct	2	Configure OSPF...
Route1	Correct	2	Configure OSPF...
Route2	Correct	2	Configure OSPF...
Passive Interface		0	Routing
GigabitEthernet0/0/0	Correct	2	Modify OSPF O...

Score : 103/103

Item Count : 51/51

Component	Items/Total	Score
Activate OSPF process	6/6	6/6
Configure OSPF Inerfaces	12/12	12/12
Configure OSPF networks	10/10	20/20
Configure Router ID	3/3	9/9
Modify OSPF Operation	18/18	48/48
Propagate default route	2/2	8/8

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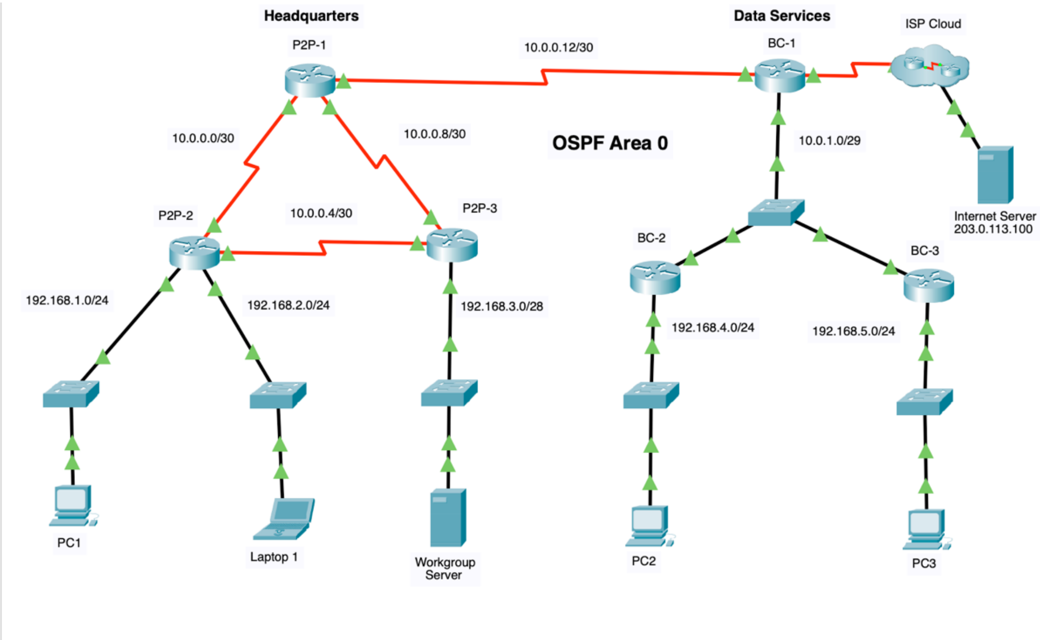

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P2P-1

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up


P2P-1>
P2P-1>
P2P-1>
P2P-1>en
P2P-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
P2P-1(config)#router ospf 10
P2P-1(config-router)#network 10.0.0.0 0.0.0.3 area 0
P2P-1(config-router)#network 10.0.0.8 0.0.0.3 area 0
P2P-1(config-router)#network 10.0.0.12 0.0.0.3 area 0
P2P-1(config-router)#
12:46:57: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.2.1 on Serial0/1/0 from LOADING to FULL, Loading Done

12:48:33: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.3.1 on Serial0/1/1 from LOADING to FULL, Loading Done

12:51:25: %OSPF-5-ADJCHG: Process 10, Nbr 64.0.100.2 on Serial0/2/0 from LOADING to FULL, Loading Done




P2P-1 con0 is now available




Press RETURN to get started.






12:55:46: %OSPF-5-ADJCHG: Process 10, Nbr 6.6.6.6 on Serial0/2/0 from LOADING to FULL, Loading Done


P2P-1>en
P2P-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
P2P-1(config)#router ospf 10
P2P-1(config-router)#auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
P2P-1(config-router)#auto-cost reference-bandwidth 1000
P2P-1(config-router)#exit
P2P-1(config)#interface Serial0/1/1
P2P-1(config-if)#ip ospf cost 50
P2P-1(config-if)#exit
P2P-1(config)#interface Serial0/2/0
P2P-1(config-if)#ip ospf hello-interval 20
P2P-1(config-if)#ip ospf dead-interval 80
P2P-1(config-if)#
13:09:16: %OSPF-5-ADJCHG: Process 10, Nbr 6.6.6.6 on Serial0/2/0 from FULL to DOWN, Neighbor Down: Dead timer expired

13:09:16: %OSPF-5-ADJCHG: Process 10, Nbr 6.6.6.6 on Serial0/2/0 from FULL to DOWN, Neighbor Down: Interface down or detached

13:09:56: %OSPF-5-ADJCHG: Process 10, Nbr 6.6.6.6 on Serial0/2/0 from LOADING to FULL, Loading Done

P2P-1(config-if)#exit
P2P-1(config)#ip ospf 10
                 ^
% Invalid input detected at '^' marker.

P2P-1(config)#exit
P2P-1#
%SYS-5-CONFIG_I: Configured from console by console

P2P-1#ip ospf 10
         ^
% Invalid input detected at '^' marker.

P2P-1#router ospf 10
             ^
% Invalid input detected at '^' marker.

P2P-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
P2P-1(config)#router ospf 10
P2P-1(config-router)#auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
P2P-1(config-router)#
```

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P2P-2

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up


P2P-2>
P2P-2>en
P2P-2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
P2P-2(config)#router ospf 10
P2P-2(config-router)#network 10.0.0.0 0.0.0.3 area 0
P2P-2(config-router)#
12:46:57: %OSPF-5-ADJCHG: Process 10, Nbr 10.0.0.13 on Serial0/1/0 from LOADING to FULL, Loading Done

P2P-2(config-router)#network 10.0.0.4 0.0.0.3 area 0
P2P-2(config-router)#network 192.168.1.0 0.0.0.255 area 0
P2P-2(config-router)#network 192.168.2.0 0.0.0.255 area 0
P2P-2(config-router)#
12:48:23: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.3.1 on Serial0/1/1 from LOADING to FULL, Loading Done


P2P-2 con0 is now available


Press RETURN to get started.


P2P-2>en
P2P-2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
P2P-2(config)#router ospf 10
P2P-2(config-router)#passive-interface GigabitEthernet0/0/0
P2P-2(config-router)#passive-interface GigabitEthernet0/0/1
P2P-2(config-router)#auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
P2P-2(config-router)#auto-cost reference-bandwidth 1000
P2P-2(config-router)#


P2P-2 con0 is now available


Press RETURN to get started.


P2P-2>en
P2P-2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
P2P-2(config)#router ospf 10
P2P-2(config-router)#auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
P2P-2(config-router)#
```

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P2P-3

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up


P2P-3>en
P2P-3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
P2P-3(config)#router ospf 10
P2P-3(config-router)#network 10.0.0.4 0.0.0.3 area 0
P2P-3(config-router)#network 10.0.0.8 0.0.0.3 area 0
12:48:23: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.2.1 on Serial0/1/0 from LOADING to FULL, Loading Done

P2P-3(config-router)#
12:48:33: %OSPF-5-ADJCHG: Process 10, Nbr 10.0.0.13 on Serial0/1/1 from LOADING to FULL, Loading Done

P2P-3(config-router)#network 192.168.3.0 0.0.0.15 area 0
P2P-3(config-router)#network 10.0.0.8 0.0.0.3 area 0
P2P-3(config-router)#




P2P-3 con0 is now available




Press RETURN to get started.








P2P-3>en
P2P-3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
P2P-3(config)#router ospf 10
P2P-3(config-router)#passive-interface GigabitEthernet0/0/0
P2P-3(config-router)#auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
P2P-3(config-router)#auto-cost reference-bandwidth 1000
P2P-3(config-router)#




P2P-3 con0 is now available




Press RETURN to get started.








P2P-3>en
P2P-3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
P2P-3(config)#router ospf 10
P2P-3(config-router)#auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
P2P-3(config-router)#
```

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BC-1

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to down

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%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up

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%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up


BC-1>en
BC-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
BC-1(config)#interface GigabitEthernet0/0/0
BC-1(config-if)#ip ospf 10 area 0
BC-1(config-if)#exit
BC-1(config)#interface Serial0/1/0
BC-1(config-if)#ip ospf 10 area 0
BC-1(config-if)#
12:51:25: %OSPF-5-ADJCHG: Process 10, Nbr 10.0.0.13 on Serial0/1/0 from LOADING to FULL, Loading Done

BC-1(config-if)#
12:52:55: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.4.1 on GigabitEthernet0/0/0 from LOADING to FULL, Loading Done

12:54:35: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.5.1 on GigabitEthernet0/0/0 from LOADING to FULL, Loading Done

BC-1(config-if)#exit
BC-1(config)#router ospf 10
BC-1(config-router)#router-id 6.6.6.6
BC-1(config-router)#
12:55:38: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.4.1 on GigabitEthernet0/0/0 from LOADING to FULL, Loading Done

12:55:46: %OSPF-5-ADJCHG: Process 10, Nbr 10.0.0.13 on Serial0/1/0 from LOADING to FULL, Loading Done

BC-1(config-router)#
BC-1(config-router)#passive-interface Serial0/1/1
BC-1(config-router)#exit
BC-1(config)#interface GigabitEthernet0/0/0
BC-1(config-if)#ip ospf priority 255
BC-1(config-if)#exit
BC-1(config)#ip route 0.0.0.0 0.0.0.0 Serial0/1/1
%Default route without gateway, if not a point-to-point interface, may impact performance
BC-1(config)#router ospf 10
BC-1(config-router)#default-information originate
BC-1(config-router)#auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
BC-1(config-router)#auto-cost reference-bandwidth 1000
BC-1(config-router)#exit
BC-1(config)#interface Serial0/1/0
BC-1(config-if)#
13:09:16: %OSPF-5-ADJCHG: Process 10, Nbr 10.0.0.13 on Serial0/1/0 from FULL to DOWN, Neighbor Down: Dead timer expired

13:09:16: %OSPF-5-ADJCHG: Process 10, Nbr 10.0.0.13 on Serial0/1/0 from FULL to DOWN, Neighbor Down: Interface down or detached

BC-1(config-if)#ip ospf hello-interval 20
BC-1(config-if)#ip ospf dead-interval 80
BC-1(config-if)#
BC-1(config-if)#
13:09:56: %OSPF-5-ADJCHG: Process 10, Nbr 10.0.0.13 on Serial0/1/0 from LOADING to FULL, Loading Done

BC-1(config-if)#
BC-1#
%SYS-5-CONFIG_I: Configured from console by console

BC-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
BC-1(config)#router ospf 10
BC-1(config-router)#auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
BC-1(config-router)#
```

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BC-2

Physical Config CLI Attributes

IOS Command Line Interface

```
GPL code under the terms of GPL Version 2.0.For more details, see the
documentation or "License Notice" file accompanying the IOS - XE software,
or the applicable URL provided on the flyer accompanying the IOS - XE
software.


This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

cisco ISR4321/K9 (1RU) processor with 1687137K/6147K bytes of memory.
Processor board ID FLM2041W2HD
2 Gigabit Ethernet interfaces
2 Serial interfaces
32768K bytes of non-volatile configuration memory.
4194304K bytes of physical memory.
3223551K bytes of flash memory at bootflash:.


Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up


BC-2>en
BC-2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
BC-2(config)#interface GigabitEthernet0/0/0
BC-2(config-if)#ip ospf 10 area 0
BC-2(config-if)#exit
BC-2(config)#interface GigabitEthernet0/0/1
BC-2(config-if)#ip ospf 10 area 0
BC-2(config-if)#
12:52:55: %OSPF-5-ADJCHG: Process 10, Nbr 64.0.100.2 on GigabitEthernet0/0/1 from LOADING to FULL, Loading Done

12:54:33: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.5.1 on GigabitEthernet0/0/1 from LOADING to FULL, Loading Done

12:55:38: %OSPF-5-ADJCHG: Process 10, Nbr 6.6.6.6 on GigabitEthernet0/0/1 from LOADING to FULL, Loading Done

BC-2(config-if)#exit
BC-2(config)#router ospf 10
BC-2(config-router)#router-id 5.5.5.5
BC-2(config-router)#passive-interface GigabitEthernet0/0/0
BC-2(config-router)#auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
BC-2(config-router)#auto-cost reference-bandwidth 1000
BC-2(config-router)#








BC-2 con0 is now available






Press RETURN to get started.













BC-2>en
BC-2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
BC-2(config)#router ospf 10
BC-2(config-router)#auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
BC-2(config-router)#
```

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BC-3

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
with ABSOLUTELY NO WARRANTY.You can redistribute and / or modify such
GPL code under the terms of GPL Version 2.0.For more details, see the
documentation or "License Notice" file accompanying the IOS - XE software,
or the applicable URL provided on the flyer accompanying the IOS - XE
software.



This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

cisco ISR4321/K9 (1RU) processor with 1687137K/6147K bytes of memory.
Processor board ID FLM2041W2HD
2 Gigabit Ethernet interfaces
2 Serial interfaces
32768K bytes of non-volatile configuration memory.
4194304K bytes of physical memory.
3223551K bytes of flash memory at bootflash:.


Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up


BC-3>
BC-3>en
BC-3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
BC-3(config)#interface GigabitEthernet0/0/0
BC-3(config-if)#ip ospf 10 area 0
BC-3(config-if)#exit
BC-3(config)#interface GigabitEthernet0/0/1
BC-3(config-if)#ip ospf 10 area 0
BC-3(config-if)#
12:54:33: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.4.1 on GigabitEthernet0/0/1 from LOADING to FULL, Loading Done

12:54:35: %OSPF-5-ADJCHG: Process 10, Nbr 64.0.100.2 on GigabitEthernet0/0/1 from LOADING to FULL, Loading Done

BC-3(config-if)#exit
BC-3(config)#router ospf 10
BC-3(config-router)#router-id 4.4.4.4
BC-3(config-router)#passive-interface GigabitEthernet0/0/0
BC-3(config-router)#auto-cost reference-bandwidth 1000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
BC-3(config-router)#auto-cost reference-bandwidth 1000
BC-3(config-router)#







BC-3 con0 is now available






Press RETURN to get started.














BC-3>en
BC-3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
BC-3(config)#router ospf 10
BC-3(config-router)#auto-cost reference-bandwidth 10000
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
BC-3(config-router)#
```

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